

Diffusion sampling quality hinges on time-step discretization

LD3: Learning to Discretize Denoising Diffusion ODEs

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- DPMs require many NFEs; discretization choices drive quality/cost.
- No universal schedule across dataset–model–solver; schedules matter.
- Claim: Learning solver- and network-aware time steps improves FID at low NFE without retraining denoisers.

2.38 FID (CIFAR10, 10 NFE)	2.27 FID (AFHQv2, 10 NFE)	5–40m Typical LD3 training (1 GPU)	7 Pretrained models evaluated
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LD3 learns discretization by minimizing global truncation error

- Teacher solver uses fine steps to approximate the diffusion ODE solution.
- Student uses $NFE \leq 10$ with learnable time steps $\{t_i\}$ via a monotone τ_ξ .
- Objective (hard): minimize $d(\Psi_\xi(x_T), \Psi^*(x_T))$; aligns with reducing $KL(p_\xi || q)$.
- Soft teacher forcing: allow x'_T within an $r\sigma_T$ -ball so $\Psi_\xi(x'_T)$ matches $\Psi^*(x_T)$; easier optimization with theory bounding KL when $L_{soft} \rightarrow 0$.

Concept figure: Teacher (dense) vs Student (few steps); hard vs soft matching.

LD3 yields consistent FID gains at low NFE across solvers and domains

- AFHQv2 @ NFE=4 (iPNDM): 9.96 vs best baseline 12.89.
- CIFAR10 @ NFE=10 (iPNDM): 2.38; FFHQ @ NFE=10 (iPNDM): 3.25.
- Latent diffusion (LSUN, ImageNet) and text-to-image (SD v1.5, InstaFlow) also improve; e.g., InstaFlow NFE=2: 22.56 → 15.49.

Single-figure placeholder: Compact summary of Table 2/4/5/6 — FID vs NFE showing LD3 vs baselines.

LD3 produces visibly cleaner samples at few steps

Qualitative placeholder: Stable Diffusion v1.5 (iPNDM). Left NFE=6, Right NFE=5. Prompts: "A red apple on a white table." / "A medieval knight standing in a castle courtyard."

Qualitative placeholder: Side-by-side across AFHQv2, ImageNet, FFHQ, LSUN-Bedroom comparing LD3 vs DMN/GITS/Uniform at same noise/solver/NFE.

- Fewer color shifts and reduced banding vs baselines.
- Sharper edges and finer textures at the same NFE.
- More consistent structure under strong guidance.

Same initial noise, solver, and NFE for fair comparison.

LD3 improves few-step diffusion sampling without network retraining

- Solver- and network-aware time schedules learned by minimizing global error.
- Consistent gains across pixel, latent, and text-to-image models.
- Fast: minutes to train with ~100 teacher pairs.

Analysis placeholder: Optimized steps outperform EDM/Uniform/LogSNR/Quadratic (see Table 3).

Efficiency placeholder: Training time vs NFE; LD3 < 1 hour on single GPU.

- Limitations: trained per NFE; requires differentiable ODE solver; may not match best distilled models at highest quality.
- Open questions: joint optimization with solver order; adaptive r scheduling; transferability across datasets/conditions.
- Broader impact: improved sampling could aid or exacerbate misuse; responsible deployment advised.